

CS667: Practical Data Science

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TBD

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Course Description

Welcome to CS667 - Practical Data Science! This project-based course is designed to guide you through the full lifecycle of a data science project, from ideation to implementation. You will explore how to formulate impactful questions, acquire and prepare data, build and validate models, and develop deployable solutions.

Our aim is to foster the mindset and technical fluency required to thrive in today's data-driven world. The course emphasizes AI-first and Generative AI-first (GenAI-first) design approaches, integrating artificial intelligence from the outset and exploring how generative models can transform system design and user interactions.

Required Materials

- Access to computing resources (laptop with Python environment)
- Course materials and notebooks available on course repository
- Python libraries: Gradio, Chainlit, various ML/AI frameworks
- Access to cloud computing platforms (details TBD)
- GitHub account for project submissions

Prerequisites/Corequisites

Prerequisites: Basic programming experience (preferably Python), fundamental knowledge of data structures and algorithms, introductory statistics. Corequisites: None.

Course Objectives

Successful students will:

1. Develop end-to-end data science workflows from problem formulation to deployment
2. Master practical tools and frameworks for building AI-powered applications
3. Create deployable, scalable, and intelligent solutions using modern AI/ML technologies
4. Implement AI-first and GenAI-first design approaches in real-world projects
5. Build interactive applications using frameworks like Gradio and Chainlit
6. Integrate various AI services and APIs into cohesive data science solutions
7. Develop skills in performance monitoring and optimization of ML systems
8. Communicate technical solutions effectively to diverse audiences

Course Structure

Class Structure

This is a project-based course emphasizing hands-on learning and practical application. Each week will combine theoretical concepts with implementation exercises, building toward comprehensive project deliverables.

Classes will include interactive coding sessions, project work time, peer reviews, and presentations of student work. A strong emphasis is placed on building deployable solutions that demonstrate real-world applicability.

Assessments

The course assessment is entirely project-based, focusing on practical deliverables that demonstrate mastery of data science workflows and AI integration.

Session Projects

Each session features hands-on projects ranging from simple applications to complex AI-powered systems. Projects build upon each other to create a comprehensive portfolio.

Major Projects

Several major projects throughout the semester, including the Chainlit MCP SQL Demo and other industry-relevant applications that showcase AI-first design principles.

Final Portfolio and Presentation

Students will compile their work into a comprehensive portfolio demonstrating their journey from basic implementations to sophisticated AI-powered solutions, culminating in a final presentation.

Grading Policy

The typical university grading scale will be used. Grades will count the assessments using the following proportions:

- 40% of your grade will be determined by session project assignments and deliverables.
- 25% of your grade will be determined by major project implementations (Chainlit MCP SQL Demo and similar).
- 15% of your grade will be determined by peer reviews and collaboration contributions.
- 10% of your grade will be determined by class participation and engagement.
- 10% of your grade will be determined by final portfolio presentation and documentation.

Course Policies

During Class

Laptops and tablets are encouraged for coding exercises and project work. Please use technology responsibly and focus on course-related activities. Collaboration is highly encouraged, but ensure individual contribution is clearly documented. Eating and drinking are allowed but should not disrupt the interactive nature of the class.

Attendance Policy

Regular attendance is essential due to the project-based and collaborative nature of this course. Valid excuses for absence will be accepted before class when possible. Excessive absences without valid excuse may significantly impact participation grades. Make-up work for project sessions will be challenging due to the iterative nature of the coursework.

Policies on Incomplete Grades and Late Assignments

Given the project-based nature of this course, timely submission is crucial as many assignments build upon previous work. Late assignments will be accepted for no penalty if a valid excuse is communicated before the deadline. After the deadline, assignments will be accepted with a 10% deduction per day up to 3 days late. After this period, assignments will receive zero credit unless extraordinary circumstances are documented.

Academic Integrity and Honesty

Students are required to comply with university policy on academic integrity. In this course, collaboration is encouraged, but individual contributions must be clearly documented. When using AI tools (ChatGPT, Copilot, etc.) for code generation or problem-solving, proper attribution is required. Plagiarism of code without attribution, especially from online sources, will not be tolerated. All external resources, APIs, and frameworks must be properly cited.

Accommodations for Disabilities

Reasonable accommodations will be made for students with verifiable disabilities. Please contact the Disability Services Office early in the semester to arrange appropriate accommodations. Alternative formats for project deliverables can be arranged as needed.

AI and Technology Use Policy

This course encourages the responsible use of AI tools as part of the learning process. Students are expected to understand and be able to explain any AI-generated code they submit. Using AI tools for learning and productivity is encouraged, but students must demonstrate understanding of the underlying concepts and be able to modify and extend AI-generated solutions independently.

Course Schedule and Learning Goals

The schedule is tentative and subject to change based on class progress and interests. Learning goals should be viewed as key concepts to master each session and serve as a study guide for project development.

Session 01 - Introduction & Gradio Hello World:

- Understand the full data science project lifecycle
- Set up development environment and tools
- Create first interactive web application using Gradio
- Explore AI-first design principles

Session 02 - Gradio Fundamentals & UI Design:

- Master Gradio interface components and layouts
- Design user-friendly AI application interfaces
- Implement basic input/output handling
- Deploy applications locally

Session 03 - Image Classification with Gradio:

- Integrate computer vision models into web applications
- Handle file uploads and image processing
- Implement model inference pipelines
- Create interactive image analysis tools

Session 04 - Advanced Image Classification:

- Optimize image classification performance
- Implement batch processing capabilities
- Add visualization and explanation features
- Handle edge cases and error management

Session 05 - Text Generation Applications:

- Integrate large language models into applications
- Design text-based AI interfaces
- Implement streaming and real-time text generation
- Handle various text input/output formats

Session 06 - Advanced Text Generation & NLP:

- Fine-tune text generation parameters
- Implement context management for conversations
- Add text analysis and preprocessing features
- Create domain-specific text applications

Session 07 - Visual Question Answering (Basic):

- Combine computer vision and natural language processing
- Implement multimodal AI applications
- Design interfaces for complex AI interactions
- Handle multiple input modalities simultaneously

Session 08 - Visual Question Answering (Advanced) & Midterm Project Review:

- Optimize multimodal model performance
- Implement advanced VQA features and capabilities
- Present midterm project portfolio
- Peer review and feedback session

Session 09 - Web Development Assistant:

- Create AI-powered development tools
- Implement code generation and assistance features
- Design developer-focused AI interfaces
- Integrate multiple AI services and APIs

Session 10 - Advanced Web Development & Integration:

- Build complex web-based AI applications
- Implement authentication and user management
- Deploy applications to cloud platforms
- Handle scalability and performance considerations

Session 11 - Performance Tracking with TrackIO:

- Implement monitoring and analytics for AI applications
- Design performance dashboards and metrics
- Track model performance and user interactions
- Optimize applications based on performance data

Session 12 - System Optimization & Monitoring:

- Advanced performance optimization techniques
- Implement A/B testing for AI features
- Design alert systems and automated monitoring
- Analyze and interpret system metrics

Session 13 - Gradio and MCP Integration:

- Understand Model Context Protocol (MCP)
- Integrate MCP with Gradio applications
- Design context-aware AI systems
- Implement advanced inter-service communication

Session 14 - Chainlit MCP SQL Demo Project:

- Implement the Chainlit MCP SQL demonstration
- Create database-integrated AI applications
- Design natural language database interfaces
- Deploy and demonstrate working applications

Session 15 - Final Projects & Integration & Portfolio Presentations:

- Complete comprehensive project portfolio
- Present final applications and demonstrations
- Integrate multiple technologies into cohesive solutions
- Demonstrate AI-first design mastery

Session 16 - Final Portfolio Submission & Course Wrap-up:

- Submit complete project portfolios
- Reflect on learning journey and skill development
- Discuss future applications and career paths
- Plan continued learning in AI and data science